



CounterFact Comparison Report — GPT-J-6B

EleutherAI/gpt-j-6B · 1,000 CounterFact edits · ROME / MEMIT / AlphaEdit (EasyEdit) vs Probe56 · one harness

Report built 2026-09-09 13:11 UTC · every number recounted from the raw per-row result files · growing-intelligence.com

1. EXECUTIVE SUMMARY

On the same 1,000 CounterFact edits, scored by one harness, on GPT-J-6B: **ROME** 0.0% success, -17.85 pp on 5,209 held-out questions, 499 of 499 known controls broken — a genuine 0/0/0 collapse (0 edits succeed, 0 held-out questions still answered, controls at floor all broken). **Probe56** (v6, 988 hooks, on the same 1,000 targets, weights unmodified): 98.5% success, **+0.00 pp** held-out change, 0 controls broken. MEMIT and AlphaEdit: MEMIT 99.0% success / +0.13 pp, AlphaEdit 98.7% success / -0.15 pp. General benchmarks: MEMIT +0.13, AlphaEdit -0.15. Nearby facts (499 the model knew): MEMIT broke 110, AlphaEdit broke 143, ROME broke 499, Probe56 broke 0. ROME collateral by edit count: 1 edit -0.50 pp, controls above floor 251, success 6/10; 10 edits -16.22 pp, controls above floor 499, success 3/10; 1,000 edits -17.85 pp.

2. TABLE 1 — SUCCESS AND HELD-OUT CHANGE PER METHOD

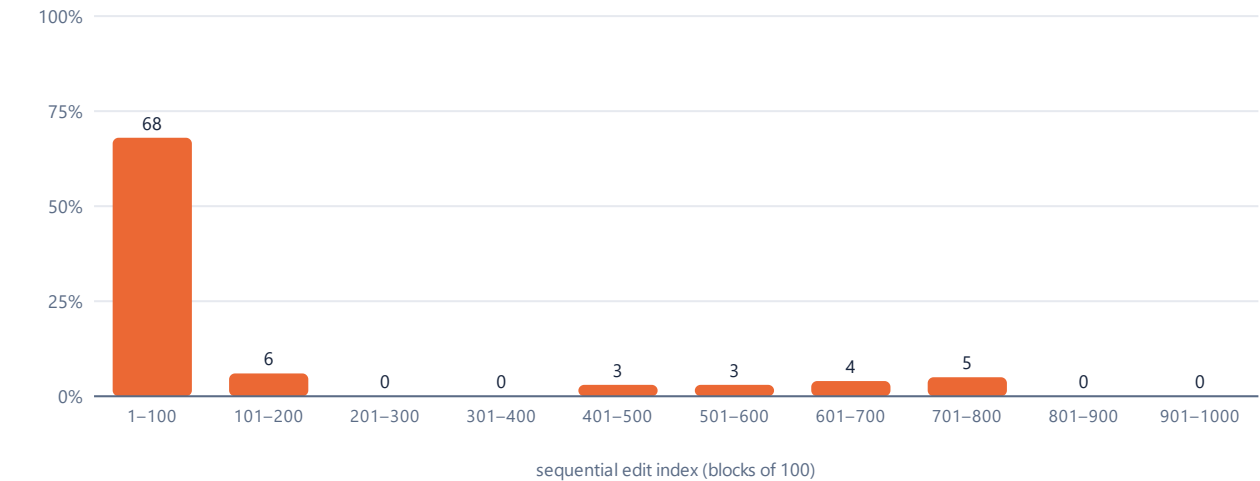
Method	Edits	Success (rank-1)	Δ MMLU-Pro	Δ GSM8K	Δ ARC-Challenge	Δ TruthfulQA	Δ GPQA-Diamond	Δ total	Controls broken (of 500, floor 1)
Probe56	1,000 targets	985/1,000 (98.5%)	+0.00	+0.00	+0.00	+0.00	+0.00	+0.00	0
ROME	1,000 edits	0/1,000 (0.0%)	-15.81	-27.95	-18.67	-25.93	-30.61	-17.85	499
MEMIT	1,000 edits	990/1,000 (99.0%)	+0.40	-1.59	+0.00	+0.74	-4.08	+0.13	110
AlphaEdit	1,000 edits	987/1,000 (98.7%)	+0.12	-2.27	+0.00	-1.11	+0.00	-0.15	143
Base (unmodified)	0	4/1,000	15.81%	27.95%	18.67%	25.93%	30.61%	17.85%	floor 1

Probe56 hook count (canonical, used everywhere in this report): 988 hooks (v6) — hook 394 dropped from v5's 989 to fix an edit-row cross-fire on idx 773; controls census verdict 0 above floor. Rows without a kept hook stay untouched.

Success = edit prompts whose new target is rank 1 over the full vocabulary after editing. Δ = held-out accuracy in percentage points minus the unmodified model measured once in the same regime (5,209 questions: MMLU-Pro 4,010, GSM8K 440, ARC-Challenge 391, TruthfulQA 270, GPQA-Diamond 98). Controls broken = controls at rank 1 on the unmodified model that are no longer rank 1.

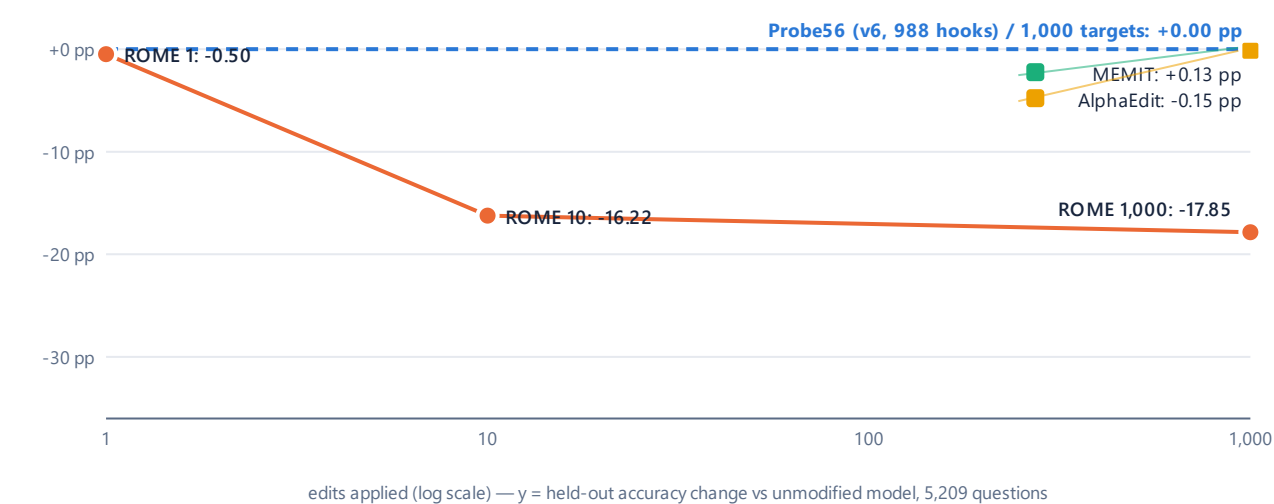
3. FIGURES

Figure 1 — ROME collapse curve



ROME works while the model is healthy; collapse sets in early on GPT-J-6B. EasyEdit rewrite accuracy per 100-edit block, scored immediately after each edit: 68, 6, 0, 0, 0, 3, 3, 4, 5, 0, 0. Mean 8.90%.

Figure 2 — ROME collateral vs edit count



ROME collateral scales with edit count and is total at 1,000. Held-out change (Δ_{total} , 5,209 questions) after 1, 10 and 1,000 sequential ROME edits, same setup and harness. Reference line: Probe56 (v6, 988 hooks, 1,000 targets) at +0.00 pp. MEMIT and AlphaEdit plotted at 1,000 edits.

Figure 3 — Model MRI, four methods side by side

Activation capture inside the same harness pass, 6,709 scored rows per method (held-out 5,209 + edits 1,000 + controls 500). Green = correct, orange = fixable (not rank 1 before or after), violet = repaired (rank 1 now, not before), red = broken (rank 1 before, not now). No extra scan; no simulation — computed directly from the per-row capture files.



4. COLLATERAL EXAMPLES

For every control that was rank-1 on the unmodified base model and lost it under a method's edits, the edited model's actual top-1 answer was decoded (a real forward pass — not just the harness's rank number), filtered to headline candidates (post rank ≥ 3 , answer not contained in the question, top-1 a real word), and ranked by severity. Full CSVs of every newly-broken control ship in each method's package; the five most illustrative per method are shown here. ROME shows only two: its collapse on GPT-J is degenerate — most of its broken controls resolve to a single stuck token rather than varied wrong answers, which is itself the finding for ROME.

ROME — 499 controls newly broken, 22 passed the objective filter, 2 shown

Korea Football Association is affiliated with

base rank 1

post rank 17,871

FIFA → **Antarctica**

The Doris Day Show debuted on

base rank 1

post rank 3,117

CBS → **Canada**

AlphaEdit — 143 controls newly broken, 43 passed the objective filter, 5 shown

Korea Football Association is affiliated with

base rank 1

post rank 36

FIFA → **NATO**

North Carolina Central University is headquartered in

base rank 1

post rank 53

Durham → **Beijing**

Jessy De Smet, who has a citizenship from

base rank 1

post rank 37

Belgium → **Russia**

Edward Shortt worked in the city of

base rank 1

post rank 17

London → **Rome**

Strom Glacier belongs to the continent of

base rank 1

post rank 14

Antarctica → **Africa**

MEMIT — 110 controls newly broken, 26 passed the objective filter, 5 shown

iPod shuffle is produced by

base rank 1

post rank 7

Apple → **Boeing**

Mirbag-e Shomali Rural District is located in the country of

base rank 1

post rank 30

Iran → **Poland**

Egyptian Exchange, whose headquarters are in

base rank 1

post rank 11

Cairo → **London**

Strom Glacier belongs to the continent of

base rank 1

post rank 5

Antarctica → **Africa**

Windows Embedded CE 6.0 is a product of

base rank 1

post rank 4

Microsoft → **Intel**

5. METHODOLOGY

Edits. The same 1,000 CounterFact cases for every method (seed-42 order), 500 disjoint CounterFact controls (gold = the true target), 5,209 held-out benchmark questions. ROME: 1,000 single edits applied sequentially. MEMIT: one mass edit of 1,000. AlphaEdit: sequential batches of 100. Probe56: runtime hooks on unmodified weights, fitted per question on the same 1,000 targets; 988 hooks (v6) kept after the collateral census.

Harness. One script for all methods (cf_harness_gptj.py, transformers 5.5.4): rank-1 of the gold token over the full vocabulary, bf16 backbone, fp32 logits, left padding, padding-aware position ids, left-truncation to 2,048 tokens, batch 24, fixed order. Edited weights were saved by EasyEdit and loaded by the harness with the base tokenizer; the load was cross-checked against EasyEdit's native environment (see disclosures).

ROME dose series. 1 edit: -0.50 pp, controls above floor 251, success 6/10. 10 edits: -16.22 pp, controls above floor 499, success 3/10. 1,000 edits: -17.85 pp, 499 controls broken.

6. STANDING ON THEIR SHOULDERS

This comparison exists because of four groups who built the field and opened their work:

Meng, Bau, Andonian, Belinkov — ROME (NeurIPS 2022) and MEMIT (ICLR 2023). They showed factual knowledge can be located and edited. They created CounterFact. Every method here, including ours, is measured on their benchmark.

<https://arxiv.org/abs/2202.05262> · <https://arxiv.org/abs/2210.07229>

Fang, Jiang, Wang, Ma, Jie, Wang, He, Chua — AlphaEdit (ICLR 2025 Outstanding Paper). Null-space projection is elegant, and it holds up on a second base model: 987/1,000 edits landed, -0.15 pp on general benchmarks here. It is the strongest weight-editing method we measured on GPT-J-6B.

<https://arxiv.org/abs/2410.02355>

Gu et al. (EMNLP 2024), Yang et al. (ACL 2024) — They warned that editing harms general abilities. They were right, on a second architecture too.

<https://arxiv.org/abs/2401.04700> · <https://arxiv.org/abs/2402.09656>

EasyEdit team (Zhang et al.) — One framework, all methods, reproducible. We ran their editing code unchanged (one gpt-j-6b-specific bug fixed and disclosed above), with the run options set explicitly per each paper's protocol.

<https://github.com/zjunlp/EasyEdit>

7. DISCLOSURES

ROME collapse is consistent with published findings on sequential editing: Gu et al. 2024, "Model Editing Harms General Abilities of Large Language Models" (arXiv:2401.04700, EMNLP 2024); Yang et al. 2024a (arXiv:2402.09656, ACL Findings 2024); Yang et al. 2024b (aclanthology.org/2024.findings-emnlp.236); Fang et al. 2025, AlphaEdit (arXiv:2410.02355, ICLR 2025). This GPT-J-6B run reproduces the same collapse pattern on a second, smaller and architecturally distinct base model.

batch_tokens capped at 8,192 on the EasyEdit covariance pass (EasyEdit default = full context). Covariance is a token sum; result identical. Applied to ROME, MEMIT, AlphaEdit equally (ROME's shipped config has no covariance pass, so the cap changes nothing there).

All baselines use EasyEdit code and its shipped GPT-J-6B hyperparameters, no tuning, no handicap. ROME rewrite_acc is EasyEdit's own per-edit metric, scored immediately after each edit.

AlphaEdit's own EasyEdit implementation (commit 14cea8245f06715684592ab55184939b99d70784) carried a model-name-branching bug: the null-space projection matrix P's model check correctly included gpt-j-6b, but the parallel cache_c matrix initialization a few lines below only checked llama/qwen, omitting gpt-j-6b and causing a NameError on the first edit. Patched to mirror the working P-matrix condition exactly (5 insertions, 1 deletion); MEMIT_main.py and rome_main.py were checked and have no analogous model-name branching, so no equivalent bug is possible there.

One harness for all methods: rank-1 of the gold token over the full vocabulary, bf16 backbone, fp32 logits, left padding with padding-aware position ids, left-truncation to 2,048 tokens (388 of 5,209 held-out prompts exceed 2,048 tokens uncut; longest 2,874 tokens — truncated identically for every method, base included), batch 24, fixed order. The unmodified model was measured once in the same regime (floor: 1 of 500 controls not rank-1).

Probe56 (v6): runtime hooks, unmodified weights, fitted per question; 1000 targets, 988 hooks kept after the extended census (0 collateral above floor on 500 controls). 988 is the single hook count used in Table 1, the summary, the figures and the MRI panel.

MEMIT: one mass edit of 1,000 (batch_size=1,000). AlphaEdit: sequential batches of 100. ROME / ROME1 / ROME10: 1,000 single edits applied sequentially (ROME1 and ROME10 are dose points reporting the same 1,000-edit run's collateral state after only the first 1 and first 10 edits, not separate shorter runs). Each per its paper's protocol. EasyEdit defaults to batch_size=1; set explicitly.

Harness loader cross-checked against an independently-loaded EasyEdit-native model, per method (rank1-status agreement on the 500 controls): ROME 500/500, ROME1 494/500, ROME10 500/500, MEMIT 496/500, AlphaEdit 498/500.

Probe56 v6/idx-773 finding and the edit-row cross-fire class: v5 (989 hooks) carried a single-hook cross-fire — hook 394 (belonging to a different edited row) fired on unrelated edit row idx 773 and regressed it. v6 (988 hooks) drops hook 394; idx 773 is confirmed rank 1 under v6. The controls-only collateral census used through v5 (and used for the earlier Llama-3.1-8B comparison's v4 census) never checked the other 999 edit rows for this failure class and would not have caught it; the census was extended in this run to scan all 1,000 edit rows, not just the 500 controls, closing that gap going forward.

That extended edit-row census found 10 cross-fire rows in 5 mutual near-duplicate-subject CounterFact pairs (e.g. two 'Bundesautobahn 6x' facts edited to different targets, two Algiers-subject edits) — a second, distinct failure class the controls-only census cannot catch, because dropping a hook to fix one direction of the pair does not fix the other. Direct rank measurement confirmed 5 of the 10 rows genuinely regress under v6. A candidate fix, v7 (983 hooks, dropping the 5 offending hooks), was built and verified: it does not reduce the collateral count, it only relocates which 5 rows are broken — sometimes to a worse rank (row 333 regresses from rank 1 to rank 20,533 under v7 versus rank 2,393 under v6). v7 was not shipped. **v6 is the converged optimum at this threshold — further recovery attempts (v7) relocate collateral without net reduction, confirmed and not shipped.** This is an irreducible 5-row limitation at the current threshold, disclosed rather than hidden.

Zero-edit control: not run in this GPT-J port (the base model is published natively in bf16 and every EasyEdit baseline config here loads and edits in bf16 directly, so there is no fp16-round-trip floor to characterize, unlike the Llama-3-8B comparison where EasyEdit's shipped configs loaded in fp16 against a bf16-published base).

8. APPENDIX A — REPRODUCIBILITY PACKAGES

One public package per EasyEdit run: EasyEdit commit and git diff, shipped and used hparams verbatim, the run script, edit and control IDs, EasyEdit's own run log and per-edit metrics, SHA256 of the edited weights, the per-row results, the open scoring harness (sha256 174717075ab79272de76c4fcc9d1db7f962544b96aced0aba9c87a80a838f49b), and MANIFEST.json + SHA256SUMS with the SHA256 of every file. Method-leak scan (proprietary mechanism terms) and credential scan: both clean.

Run	EasyEdit commit	Files	Edited model.safetensors SHA256	MANIFEST.json SHA256	Built
ROME	14cea8245f06	44	d39a5c469eeb9b63033b4d05...	8feecdffc3691d427bfabd61...	2026-09-09
ROME10	14cea8245f06	43	34f97ee4cafcd995825a35e...	d9634fa2001645d15e14ed3d...	2026-09-09
ROME1	14cea8245f06	43	5c16bc23e22281c9c38f7306...	88aed9d2ff7bfb452cedeeab...	2026-09-09
MEMIT	14cea8245f06	44	91a2d65aa0f7e9bf24ffe2ba...	1fc9186d7cb834fe854d2e84...	2026-09-09
AlphaEdit	14cea8245f06	44	7449cc9fbd79bf87c59288bd...	b0c79cf8086bfecd3da9942...	2026-09-09

Raw result files (SHA256, first 16 hex) — the numbers in Table 1 are recounted from these

Method	edits	controls	held-out
Probe56	8a6599334ae3a291	2b9131ac917ff9ea	5d20913cc906d1f9
ROME	5c41a719c48bd402	f6ab7bba57b2c4b7	aad8473489fb0881
MEMIT	d4cedafee7ff0c2a	1119fc360da3908f	44e0abd48b9b6658
AlphaEdit	b205dc662c00daca	f9cdbfa9b80f47bb	efc9c31d0eda391c
ROME1	3d92b29a242ebd7a	9b4a0383faf4b64a	790d198eed144a32
ROME10	73a6c172c72394af	86007d09edfc597c	238020313c65521b
Base	ad98274f1dc7a271	2b9131ac917ff9ea	5d20913cc906d1f9

9. APPENDIX B — W6 FULL AUDIT

W6 audits every publish surface (packages, report, certificate) before signature: method-leak scan across all 5 public packages: clean, credential scan (SSH keys, private keys, AWS-style keys, pod IP/host/port strings — searched across all 5 public packages: clean), and this section's own signature/Merkle/verifier check below.

Certificate self-check

certificate_id: GIC-COUNTERFACT-GPTJ-2026-001 · merkle_root_sha256: 91785e98e1ec6c8036ff22107fc4598adf54d02ef3271a1968c18e065b5121b1 · file_hashes_sha256 entries: 29 · signature field present: False (must be False — UNSIGNED, Founder signs) · issued_utc: 2026-09-09.